

University of Sindh, Jamshoro
Institute of Mathematics & Computer Science
Department: Computer Science & Engineering (CSME)

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Topic: PEAS Framework and Task Environment Analysis

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Selected System: Option H - An AI Tutor for Personalized Student Learning Recommendations

Task 1: PEAS Specification

The AI Tutor system is designed to provide personalized learning recommendations and adaptive educational content to students across various subjects and learning levels.

Performance Measures:

- **Student Learning Gain:** Measured by improvement in assignment scores and assessments. Success metric: 20%+ score improvement over 4 weeks.
- **Personalization Accuracy:** System's ability to correctly identify student's learning style and pace. Tracked through engagement metrics and assessment alignment.
- **Time-to-Mastery:** Number of hours/sessions required for student to reach 80% proficiency on a concept. Target: reduce by 30% compared to traditional tutoring.
- **Student Engagement Rate:** Measured by active interaction time, number of questions asked, and session frequency. Target: minimum 3 hours per week engagement.

Environment:

The AI Tutor operates within an online learning platform accessible via web and mobile applications. The environment includes diverse student populations with varying prior knowledge levels, different subject domains (Mathematics, Science, Languages, History), and asynchronous interaction patterns. Key external factors include student availability, internet connectivity, device capabilities, and real-time performance data streams.

Actuators:

- **Adaptive Lesson Generator** - Creates customized content based on student profile
- **Difficulty Level Controller** - Adjusts problem complexity dynamically
- **Hint and Clue Provision System** - Delivers scaffolded support without revealing answers
- **Performance Feedback Dashboard** - Generates real-time progress reports
- **Next-Topic Recommendation Engine** - Suggests learning paths based on mastery level

Sensors:

- Answer Correctness Tracker - Records right/wrong answers and response patterns
- Time-Spent-Per-Question Monitor - Tracks cognitive load and struggling points
- Engagement Analytics - Monitors click patterns, scrolling behavior, and session duration
- Assessment Score History - Maintains longitudinal performance data
- Student Demographic and Preference Data - Age, grade level, preferred language, learning modality

Task 2: Environment Classification

Dimension	Classification	Justification
Observability	Partially Observable	Tutor cannot directly observe cognitive processes; must infer from behavior and answers.
Agents	Multi-agent	Multiple students interact simultaneously; teachers and parents influence learning.
Determinism	Stochastic	Same student on same topic shows varied performance due to mood, concentration, and distraction.
Episodicity	Sequential	Today's lesson affects tomorrow's learning path; prior knowledge is essential.
Dynamism	Dynamic	Student knowledge continuously improves; system state changes during interactions.
Granularity	Discrete	Questions are discrete units; proficiency measured on continuous 0-100% scale.

Task 3: Critical Analysis

1. Greatest Technical Challenge:

The **Partially Observable** nature presents the greatest challenge. The AI Tutor cannot directly perceive actual understanding. A student may submit correct answers through guessing, memorization, or cheating, while incorrect answers may result from careless mistakes rather than conceptual gaps. Distinguishing these scenarios requires sophisticated pattern recognition—analyzing response time distributions, answer consistency, and difficulty progression. This dual risk (false positives and false negatives) necessitates probabilistic inference using Bayesian networks or Hidden Markov Models to infer true learning states from incomplete observations.

2. Utility Function and Trade-off Analysis:

A fundamental trade-off exists between **Learning Quality** and **Student Engagement**. **Proposed Function:** $U = 0.6 \times \text{Learning_Gain} + 0.4 \times \text{Student_Engagement}$ **If we double Learning_Gain weight (0.8 vs 0.2):** The system prioritizes rigorous, challenging content. Consequences: Higher learning for persistent students, but increased frustration and dropout rates. Many students abandon the system, reducing overall engagement and accessibility.

Task 4: Structured Representation (JSON)

```
{
  "system_name": "AI Tutor for Personalized Student Learning",
  "peas": {
    "performance_measures": [
      "Student learning gain (improvement in assessment scores)",
      "Personalization accuracy (alignment with learning style)",
      "Time-to-mastery (hours needed to understand topic)",
      "Student engagement rate (interaction time and question frequency)"
    ],
    "environment": "Online educational platform with diverse student populations, varying prior knowledge levels, subject difficulty levels, and real-time performance feedback requirements",
    "actuators": [
      "Adaptive lesson generator",
      "Difficulty level controller",
      "Hint and clue provision system",
      "Performance feedback dashboard",
      "Next-topic recommendation engine"
    ],
    "sensors": [
      "Student answer correctness tracker",
      "Time-spent-per-question monitor",
      "Student engagement analytics",
      "Previous assessment score history",
      "Student demographic and preference data"
    ]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "Tutor cannot directly observe cognitive processes; must infer understanding from external behavior and answers"
    },
    "agents": {
      "choice": "Multi-agent",
      "justification": "Multiple students interact simultaneously; teachers and parents also influence the system"
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    "determinism": {
      "choice": "Stochastic",
      "justification": "Same student shows varied performance due to mood, concentration, and external factors"
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    "episodicity": {
      "choice": "Sequential",
      "justification": "Today's lesson affects tomorrow's learning path; prior knowledge is essential"
    },
    "dynamism": {
      "choice": "Dynamic",
      "justification": "Student knowledge continuously improves; system state changes during interactions"
    }
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}
```

```
    },
    "granularity": {
      "choice": "Discrete",
      "justification": "Questions are discrete units; proficiency measured on continuous 0-100% scale"
    }
  },
  "utility_function": "U = 0.6*Learning_Gain + 0.4*Student_Engagement"
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```

Submitted on: June 14, 2026